

## ELMIRA CORNING, NY, USA

WMO: 725156

Lat: 42.159N

Lon: 76.892W

Elev: 291

StdP: 97.88

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14748

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.2      | -15.5 | -22.9                         | 0.5 | -17.1 | -20.2 | 0.6 | -14.1 | 11.2                  | -1.1 | 9.8 | -3.4 | 1.7                   | 250  | 0.517 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.1                   | 32.0            | 22.0 | 30.3 | 21.3 | 28.8 | 20.6 | 23.7                | 29.1 | 22.8 | 27.9 | 21.9 | 26.8 | 4.3                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.1                            | 17.4 | 26.4 | 21.2 | 16.4 | 25.4 | 20.3 | 15.6 | 24.6 | 72.1          | 29.2 | 68.6 | 27.9 | 65.6 | 26.7 | 29.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.0           | 8.0  | 7.1 | DB                         | -23.0 | 34.7               | 3.4 | 1.9                                                | -25.4 | 36.1       | -27.5 | 37.1       | -29.4 | 38.2       | -31.9 | 39.5 | (4) |
| (5)               |      |     | WB                         | -23.2 | 25.5               | 3.4 | 1.1                                                | -25.7 | 26.3       | -27.6 | 26.9       | -29.5 | 27.6       | -32.0 | 28.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.9         | -4.2  | -3.0  | 1.3   | 7.8   | 14.1  | 18.9  | 21.2  | 20.4  | 16.6  | 10.2  | 4.0   | -0.8  | (6)  |  |
|                                                                        | DBStd       | 10.13       | 6.45  | 5.86  | 5.71  | 4.84  | 4.61  | 3.60  | 3.04  | 2.96  | 3.96  | 4.53  | 4.72  | 5.12  | (7)  |  |
|                                                                        | HDD10.0     | 1773        | 442   | 364   | 275   | 98    | 12    | 0     | 0     | 0     | 2     | 55    | 189   | 337   | (8)  |  |
|                                                                        | HDD18.3     | 3707        | 699   | 597   | 527   | 317   | 147   | 36    | 9     | 13    | 79    | 256   | 431   | 594   | (9)  |  |
|                                                                        | CDD10.0     | 1386        | 1     | 1     | 6     | 32    | 139   | 268   | 347   | 323   | 200   | 60    | 8     | 1     | (10) |  |
|                                                                        | CDD18.3     | 276         | 0     | 0     | 0     | 1     | 16    | 54    | 97    | 78    | 27    | 3     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 2878        | 0     | 0     | 6     | 48    | 245   | 578   | 973   | 726   | 272   | 30    | 1     | 0     | (12) |  |
|                                                                        | CDH26.7     | 903         | 0     | 0     | 1     | 12    | 72    | 178   | 337   | 221   | 79    | 3     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.6         | 3.0   | 3.1   | 3.2   | 3.2   | 2.7   | 2.3   | 2.1   | 2.0   | 2.1   | 2.5   | 2.7   | 2.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 869         | 51    | 47    | 66    | 75    | 82    | 98    | 85    | 80    | 87    | 68    | 74    | 57    | (15) |  |
|                                                                        | PrecMax     | 1192        | 125   | 110   | 178   | 221   | 179   | 280   | 218   | 226   | 206   | 176   | 144   | 117   | (16) |  |
|                                                                        | PrecMin     | 588         | 10    | 2     | 5     | 22    | 23    | 20    | 27    | 25    | 12    | 4     | 23    | 20    | (17) |  |
|                                                                        | PrecStd     | 156         | 28    | 24    | 35    | 38    | 39    | 49    | 45    | 45    | 45    | 38    | 34    | 26    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 15.8  | 16.4  | 23.0  | 28.2  | 31.4  | 32.9  | 34.1  | 33.2  | 32.0  | 26.6  | 20.1  | 17.1  | (19) |  |
|                                                                        |             | MCWB        | 12.7  | 10.3  | 14.4  | 16.6  | 20.6  | 23.1  | 23.0  | 22.5  | 21.0  | 18.9  | 14.3  | 12.9  | (20) |  |
|                                                                        | 2%          | DB          | 11.0  | 11.3  | 17.7  | 24.0  | 28.7  | 30.8  | 32.0  | 31.1  | 29.0  | 23.5  | 17.3  | 12.2  | (21) |  |
|                                                                        |             | MCWB        | 8.2   | 6.8   | 10.8  | 14.2  | 19.0  | 21.6  | 22.4  | 21.7  | 20.5  | 17.2  | 12.7  | 9.4   | (22) |  |
|                                                                        | 5%          | DB          | 7.3   | 7.9   | 13.8  | 20.4  | 26.2  | 28.7  | 30.2  | 29.0  | 26.4  | 21.1  | 15.1  | 9.1   | (23) |  |
|                                                                        |             | MCWB        | 5.2   | 4.5   | 8.7   | 12.3  | 17.8  | 20.7  | 21.7  | 20.9  | 19.2  | 15.6  | 11.1  | 6.8   | (24) |  |
|                                                                        | 10%         | DB          | 4.5   | 5.3   | 10.8  | 17.5  | 23.4  | 27.0  | 28.4  | 27.4  | 24.0  | 18.3  | 12.5  | 6.8   | (25) |  |
|                                                                        |             | MCWB        | 2.5   | 2.5   | 6.8   | 10.9  | 16.3  | 19.7  | 20.8  | 20.3  | 18.3  | 14.4  | 9.2   | 4.5   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 13.2  | 11.2  | 15.4  | 18.0  | 22.2  | 24.4  | 25.2  | 24.5  | 23.4  | 20.7  | 15.7  | 13.5  | (27) |  |
|                                                                        |             | MCDB        | 15.3  | 14.6  | 21.7  | 25.6  | 29.2  | 29.9  | 31.4  | 30.1  | 28.8  | 24.4  | 18.5  | 16.4  | (28) |  |
|                                                                        | 2%          | WB          | 8.4   | 7.4   | 12.0  | 15.5  | 20.4  | 23.0  | 23.9  | 23.1  | 21.8  | 18.6  | 13.8  | 9.7   | (29) |  |
|                                                                        |             | MCDB        | 10.1  | 10.7  | 15.9  | 21.2  | 26.3  | 28.5  | 29.4  | 27.8  | 26.7  | 22.2  | 16.7  | 11.6  | (30) |  |
|                                                                        | 5%          | WB          | 5.4   | 5.0   | 9.5   | 13.8  | 19.0  | 21.9  | 22.9  | 22.2  | 20.6  | 16.6  | 11.7  | 7.1   | (31) |  |
|                                                                        |             | MCDB        | 7.4   | 7.7   | 13.8  | 19.2  | 24.1  | 27.0  | 28.0  | 26.9  | 24.3  | 19.7  | 14.3  | 8.9   | (32) |  |
|                                                                        | 10%         | WB          | 2.8   | 2.8   | 6.8   | 11.7  | 17.5  | 20.8  | 22.0  | 21.4  | 19.4  | 14.7  | 9.4   | 4.6   | (33) |  |
|                                                                        |             | MCDB        | 4.4   | 5.2   | 10.1  | 16.1  | 22.4  | 25.0  | 26.7  | 25.7  | 22.8  | 18.1  | 12.4  | 6.6   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.5   | 10.3  | 11.2  | 13.2  | 13.7  | 12.9  | 13.1  | 12.8  | 12.8  | 11.9  | 10.4  | 8.4   | (35) |  |
|                                                                        |             | MCDBR       | 12.0  | 14.2  | 17.0  | 19.6  | 17.6  | 15.6  | 15.1  | 14.8  | 15.7  | 16.3  | 15.1  | 11.8  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 10.1  | 10.6  | 11.0  | 11.0  | 9.0   | 7.7   | 6.7   | 6.8   | 7.9   | 9.9   | 10.5  | 9.5   | (37) |  |
|                                                                        |             | MCWBR       | 11.5  | 12.8  | 15.6  | 16.3  | 15.0  | 13.6  | 12.9  | 12.5  | 12.4  | 13.3  | 12.8  | 10.8  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.304 | 0.322 | 0.351 | 0.370 | 0.397 | 0.445 | 0.451 | 0.441 | 0.368 | 0.357 | 0.314 | 0.295 | (40) |  |
|                                                                        |             | taud        | 2.366 | 2.318 | 2.329 | 2.339 | 2.300 | 2.189 | 2.183 | 2.199 | 2.389 | 2.417 | 2.495 | 2.460 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 835   | 877   | 892   | 898   | 878   | 833   | 823   | 819   | 864   | 828   | 825   | 820   | (42) |  |
|                                                                        |             | Edh at Noon | 82    | 101   | 113   | 120   | 129   | 144   | 144   | 137   | 105   | 91    | 72    | 69    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.46  | 2.32  | 3.37  | 4.42  | 5.22  | 5.57  | 5.69  | 5.07  | 4.07  | 2.60  | 1.69  | 1.17  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.12  | 0.29  | 0.32  | 0.50  | 0.53  | 0.50  | 0.42  | 0.30  | 0.44  | 0.28  | 0.22  | 0.17  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -116    | +101    | +39     |
| (47) Regional (30 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +95     | N/A     |

Nomenclature: See separate page